MATTHEW T. BECK

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EDUCATION

Stony Brook University — Stony Brook, NY
Ph.D., Computer Science — Expected 2031

University of Virginia — Charlottesville, VA
M.S., Data Science — May 2025

University of Virginia — Charlottesville, VA
B.S. with High Distinction, Computer Science — May 2024
Applied Mathematics and Data Science Minors, Cybersecurity Focal Path

TEACHING EXPERIENCE

Stony Brook University, Department of Computer Science — Stony Brook, NY
Teaching Assistant, Information Systems Course — Spring 2025
"Systems Administration" (Ahmad Esmaili)
- Manage weekly office hours
- Review assignments, providing feedback and grade
- Review and edit lab/homework assignments

Stony Brook University, Department of Computer Science — Stony Brook, NY
Teaching Assistant, Information Systems Course — Fall 2025
"Introduction to Network Administration" (Ahmad Esmaili)
- Manage weekly lab sessions
- Review assignments, providing feedback and grade
- Review and edit lab/homework assignments

University of Virginia, Department of Computer Science — Charlottesville, VA
Teaching Assistant, Cybersecurity course — Fall 2024
"Defense Against the Dark Arts" (Professor Wajih Ul Hassan)
- Manage weekly office hours
- Review assignments, providing feedback and grade

University of Virginia, School of Data Science — Charlottesville, VA
Instructional Assistant, Machine Learning course — Fall 2024
"Foundations of Machine Learning" (Professor Terence Johnson)
- Manage weekly office hours
- Review lab assignments, providing feedback and grade

University of Virginia, Department of Applied Mathematics — Charlottesville, VA
Teaching Assistant, Probability course — Fall 2022 - Fall 2023
"Probability" (Professor Christian Hellings and Professor Jundong Li)

- Provided in-class coaching and support
- Reviewed assignments, provided feedback and graded

RESEARCH EXPERIENCE

Stony Brook University, Department of Computer Science — Stony Brook, NY
Research Assistant, Professor R. Sekar — August 2025 - Present
- Developing complete intrusion detection system pipeline
- Building an efficient, information rich Windows log producer for intrusion detection systems

University of Virginia, Department of Computer Science — Charlottesville, VA
Research Assistant, Professor Wajih Ul Hassan — June 2024 - May 2025
- Investigating the optimization of intrusion detection systems' throughput

University of Virginia, School of Data Science — Charlottesville, VA
Team Member, Data Science Capstone — October 2024 - May 2025
- Constructing computer vision software to digitize surgical flowsheets for developing countries

PROFESSIONAL EXPERIENCE

Nasuni, Data Path Team — Boston, MA
Software Engineering Intern — Summer 2023
- Reinforced typing into python codebase through type-hinting
- Reconstructed and maintained features of the codebase to optimize the workflow of the team

LEADERSHIP EXPERIENCE

The Clover Project — Poughkeepsie, NY
CEO and Co-Founder — March 2019 - Present
- Manage over 150 student volunteers with the mission of reducing local food insecurity
- Produced over three tons of fresh produce for donation to local food banks

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++ (Advanced), SQL (Advanced), R (Advanced), C (Advanced), JavaScript (Intermediate), Java (Intermediate)
Web Development: HTML5, CSS3, Bootstrap, Node.js
Software and Tools: Docker, AWS, GCP, Azure, Git, Agile methodologies
Databases: MySQL, MongoDB, PostgreSQL
Data Science and Machine Learning: Pandas, Numpy, Scikit-learn, TensorFlow, Jupyter, Regression, Clustering, Classification, Deep Learning
Cybersecurity: Wireshark, Metasploit, Encryption, Vulnerability Analysis, Intrusion Detection, ELK stack (ElasticSearch, Logstash, Kibana), SIGMA rules, Autopsy, Sleuth Kit

PROFESSIONAL DEVELOPMENT

Cornell University Ithaca, NY
Decision Analysis Certificate 2024

- Using an evidence-based approach to model decision making processes

REFERENCES

Wajih Ul Hassan, Ph.D.
Assistant Professor
Department of Computer Science
School of Data Science
University of Virginia
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Prince Afriyie, Ph.D.
Associate Professor
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